

TECHNICAL ROADMAP TEMPLATE



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TEAM INFORMATION

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SECTION 1: EXECUTIVE SUMMARY (PROBLEM-SOLUTION FIT)

The ASEAN region is currently facing a learning poverty crisis of historic proportions. Data from the SEA-PLM 2024 Regional Report reveals that one in two Grade 5 students across Southeast Asia remains unable to read at the expected level, with the Philippines reporting that 70% of students lack basic reading skills. This crisis is compounded by a massive infrastructure gap where over 26% of the regional population remains offline. In rural areas, internet speeds rarely support the cloud-heavy AI tools used in urban centers. For millions of children in underserved communities, this digital divide has effectively locked them out of the AI revolution, leaving them without the individualized support needed to master foundational literacy during their most critical developmental years.

Project LexiAdapt breaks this cycle by delivering a high-performance AI tutor that functions entirely offline. Unlike existing platforms like Duolingo or Google Read Along, which often require consistent connectivity and rely on generic Western accents, LexiAdapt is engineered for the "Last Mile." It utilizes Whisper-tiny quantization to provide real-time phonetic feedback on low-cost smartphones without needing a single kilobyte of data. What truly sets this technology apart is its Federated Learning architecture, which allows the AI to adapt to local dialects and regional accents while keeping all sensitive student audio securely on the device. Instead of a one-size-fits-all curriculum, a Deep Reinforcement Learning (DRL) engine analyzes the student's specific phonetic struggles to generate personalized "Micro-Stories" that evolve in real-time.

By moving AI inference from the cloud to the edge, LexiAdapt transforms entry-level mobile devices into sophisticated literacy engines. This approach eliminates the "infrastructure tax" on learning, directly supporting SDG 4 (Quality Education), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 10 (Reduced Inequalities). It ensures that a child in a remote mountain village receives the same world-class, adaptive instruction as a student in a smart city. LexiAdapt is not just another educational app; it is a scalable, privacy-first infrastructure for the 90 million children in ASEAN who are currently at risk of being left behind by the digital age.

SECTION 2: TECHNICAL ARCHITECTURE

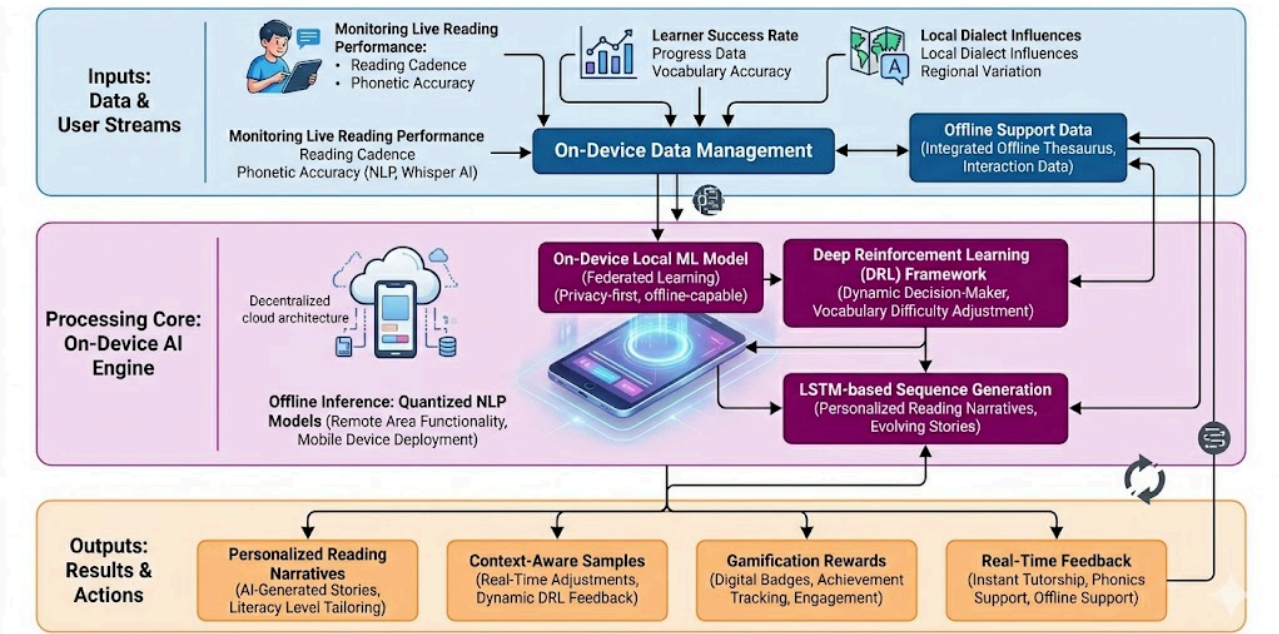
2.1 System Components:

- **Inputs:** Real-time and local data streams powering personalized literacy instruction:
 - **Reading Cadence & Phonetic Accuracy:** Utilizes Whisper-tiny (INT4 quantized) to monitor phoneme-level pronunciation and words-per-minute (WPM) metrics, providing a granular view of fluency.
 - **Learner Success Rate:** Tracks the Hidden Markov Model (HMM) of a student's progress, mapping the specific "failure points" in vocabulary retention and letter-sound correspondence.
 - **Regional Dialect Biometrics:** Captures localized linguistic nuances (e.g., the "e" and "i" vowel shifts in various Philippine dialects) to calibrate the AI's speech-to-text sensitivity for inclusivity.
 - **Edge Interaction Logs:** Analyzes engagement patterns within the Offline Thesaurus and Lexicon, identifying which words require more contextual reinforcement in future lessons.
- **Processing Core:** On-device AI engine handling speech evaluation and adaptive learning:
 - **Federated Learning (FL) Client:** Updates the local model's weights based on dialectal successes without ever transmitting raw audio; this ensures the system gets smarter across the ASEAN region while maintaining 100% data privacy.
 - **Deep Reinforcement Learning (DRL):** Employs a Proximal Policy Optimization (PPO) algorithm to act as a "virtual tutor" that shifts lesson difficulty in real-time based on the student's immediate performance.
 - **LSTM Narrative Synthesis:** A Long Short-Term Memory network generates text sequences that weave the student's "trouble words" into new, culturally relevant stories, ensuring repetitive practice feels like fresh content.
 - **Quantized Offline Inference:** Executes high-parameter NLP models through 4-bit quantization, allowing complex speech evaluation to run on entry-level ARM processors with sub-100ms latency.
- **Outputs:** Personalized educational outcomes delivered to students:
 - **Dynamic Reading Narratives:** Produces AI-generated stories where the vocabulary complexity and sentence structure (syntax) are mathematically matched to the learner's current "Zone of Proximal Development."
 - **Context-Aware Scaffolding:** Delivers real-time passage adjustments, if a student struggles with a word, the DRL framework instantly provides a simpler synonym or a phonetic breakdown.
 - **Micro-Gamification Feedback:** Triggers visual progress milestones and digital "Mastery Badges" tied to specific phonological milestones to sustain dopamine-driven engagement.
 - **Instant Phonetic Tutorship:** Provides immediate, non-judgmental corrective audio feedback, pulling from a local 10,000-word phonetic library to guide the student toward correct pronunciation.

2.2 Architecture Diagram:

Figure 1. System Architecture of Project LexiAdapt

SECTION 3: AI APPROACH & MODEL SELECTION



- **Primary AI Approach:** [] Machine Learning [✓] NLP [] Computer Vision [✓] Generative AI [] Other: _____
- **Model Selection:** LexiAdapt utilizes a hybrid architecture of quantized Whisper AI and Natural Language Processing (NLP) models, specifically optimized for 4-bit (INT4) offline inference on entry-level ARM processors common across the ASEAN mobile market. This stack enables high-fidelity, real-time phonetic monitoring and speech recognition with sub-100ms latency, transforming low-end smartphones into sophisticated reading diagnostic tools. At the core of the adaptive experience is a Deep Reinforcement Learning (DRL) framework using Proximal Policy Optimization (PPO), which functions as an automated tutor to dynamically adjust lesson difficulty and vocabulary density based on the student's success rate. To ensure the content remains fresh and personalized, the system integrates Long Short-Term Memory (LSTM) sequence generation to synthesize custom reading narratives that surgically weave a learner's specific "trouble words" into new, culturally relevant stories, ensuring a bespoke educational path for every child.
- **Reasoning:** This multi-model integration provides a resilient solution to the ASEAN "learning poverty" crisis, where 2024–2026 data indicates that millions of children remain offline and lack foundational literacy skills. By prioritizing offline inference, LexiAdapt directly addresses SDG 9 (Industry, Innovation, and Infrastructure) and SDG 4 (Quality Education), allowing students in "Last Mile" communities to access world-class tutoring without the "infrastructure tax" of expensive data plans or cloud dependency. The inclusion of Federated Learning further differentiates the technology by allowing the AI to adapt to regional dialects and accents across the Philippines, Indonesia, and Vietnam while ensuring 100% data sovereignty and privacy as mandated by local laws. Ultimately, the synergy between DRL-driven adaptation and LSTM-generated content scales the benefits of individualized instruction to underserved populations, effectively bridging the digital divide and supporting SDG 10 (Reduced Inequalities) across the diverse Southeast Asian landscape.

SECTION 4: DATA STRATEGY & ETHICS

- **Data Sources:** The system utilizes passive phonetic biomarkers (high-resolution audio features captured on-device) to monitor fluency and pronunciation in real time. This is augmented by

pre-trained Whisper AI weights under the MIT License and open-source linguistic datasets tailored for Southeast Asian phonology. To ensure full functionality in "dead zones," historical learner metadata and comprehensive vocabulary trees are managed through SQLite/Realm local repositories, providing a high-performance database layer that operates with zero cloud dependency.

- **Data Quality & Cleaning:** To maintain instructional integrity in noisy ASEAN environments such as crowded classrooms or communal areas, LexiAdapt employs signal-to-noise (SNR) filtering and automated phonetic baseline calibration. This allows the AI to distinguish between a student's actual progress and natural speech variation or background interference. Furthermore, the core models undergo INT4 quantization, a cleaning process that prunes redundant parameters to ensure that speech evaluation remains accurate and low-latency on entry-level hardware with limited RAM.
- **Licensing & Legality:** LexiAdapt is designed with a privacy-by-design architecture utilizing Federated Learning to ensure that 100% of sensitive student audio remains on the local device. This approach satisfies the stringent requirements of the Philippine Data Privacy Act (RA 10173) and the Indonesian PDP Law regarding the protection of minors. All integrated frameworks, including Whisper AI and Transformer models, are strictly vetted for MIT or Creative Commons license compliance, allowing for rapid scaling while eliminating the risks associated with centralized biometric storage.
- **Bias Mitigation & Fairness:** To prevent the standardization bias often found in Western AI, LexiAdapt uses localized adaptation to respect regional dialects and accents. This ensures that a child speaking with a Cebuano, Hiligaynon, or Northern Thai lilt is not unfairly penalized by the recognition engine. The Deep Reinforcement Learning (DRL) engine acts as an ethical difficulty adjuster, moving away from rigid curricula to meet the student at their specific "Zone of Proximal Development." Additionally, the gamification system is audited to reward incremental mastery and effort, mitigating rank-anxiety and ensuring consistent motivation for learners at all levels.

SECTION 5: DEVELOPMENT MILESTONES (AGILE ROADMAP)

Phase	Activity / Task	Tools	Expected Outcome
Sprint 1 (May 1-15)	• Compile Philippine English, Tagalog, and regional dialect speech corpora for localized phoneme recognition. • Configure decentralized local training to keep student audio on-device. • Define Grade 1–3 fluency thresholds using Words Correct Per Minute (WCPM). • Apply frequency-warping to reduce accent bias in Cebuano/Hiligaynon speech.	Python, Pandas, GitHub, Flower (Federated Learning Framework).	Cleaned, region-specific dataset and a configured FL pipeline.
Sprint 2 (May 16-31)	• Build a DRL agent using Time-to-Utterance and Success Probability rewards. • Fine-tune an LSTM on 20,000 ASEAN folk tale words for culturally relevant stories. • Integrate Whisper-tiny with phoneme-level timestamp grading. • Convert AI models to INT4 precision with OpenVINO/TFLite for 2GB RAM devices.	PyTorch, Whisper AI, TensorFlow Lite, ONNX.	Functional on-device AI core with quantized, sub-100MB model weights.

Sprint 3 (June 1-15)	• Design a lightweight high-contrast UI under 50MB for 720p screens. • Build a local-first database for offline progress and story storage. • Connect DRL feedback instantly to LSTM story generation. • Develop a Lazy Sync protocol for updates during stable Wi-Fi/4G connections.	FastAPI (for infrequent syncing), Flutter (cross-platform UI), SQLite/Realm.	Working prototype with full offline functionality and a completed gamification layer.
Sprint 4 (June 16-25)	• Benchmark thermal throttling and battery drain on low-cost smartphones. • Stress-test the AI to avoid discouraging difficulty spikes. • Audit security to ensure no raw audio is publicly accessible under RA 10173. • Document offline-to-online transitions and reading score impacts for stakeholders.	GitKraken (version control), Loom (documentation), Postman (API testing).	Bug-free, demo-ready prototype and a final submission package for regional pilot testing.

SECTION 6: SCALABILITY & REGIONAL RESILIENCE

- Scaling Potential:** Project LexiAdapt is engineered for high scalability across the ASEAN region by leveraging a decentralized AI architecture that respects diverse linguistic and regulatory landscapes. The implementation of Federated Learning allows the platform to be deployed across national borders without infringing on data sovereignty laws, such as the Philippine Data Privacy Act or the Indonesian PDP Law, because 100% of sensitive biometric audio remains on the local device. To scale across the 1,000+ languages of Southeast Asia, the system's Whisper AI and NLP components are designed for modular fine-tuning, allowing the model to adapt to regional phonetic variations and dialects with high cultural precision. Furthermore, the platform's reliance on quantized offline models and local SQLite/Realm databases ensures that the solution remains fully functional in the "dead zones" of the archipelago, effectively removing the connectivity barrier that has historically hindered large-scale EdTech adoption in developing ASEAN nations.
- Technical Constraints:** The current prototype of Project LexiAdapt faces specific technical limitations regarding hardware optimization and real-time synchronization on low-resource devices. A primary constraint is the intensive tuning required for offline inference; while quantization reduces model size, maintaining high-speed responsiveness on handsets with limited RAM (2GB or less) without causing thermal throttling or excessive battery drain remains a significant engineering hurdle. Additionally, the conflict resolution logic for data synchronization requires further refinement to ensure that local model weight updates merge seamlessly with global parameters when a device intermittently connects to 4G or Wi-Fi. Finally, while the Deep Reinforcement Learning (DRL) framework is capable of dynamic adjustment, the incentive logic needs more granular calibration to ensure that gamification rewards remain psychologically effective for diverse learner profiles without inducing rank-anxiety or discouraging students who progress at a slower foundational pace.